

ALLOC-X

TRUSTED AUTONOMOUS TREASURY

TECHNICAL FOUNDATION

A production blueprint for an autonomous treasury agent that can plan, authorize, execute, validate and prove organizational payments on XRPL.

PRODUCT SURFACES

7

Figma-ready screens

SECURITY GATES

9

Ordered before signing

DECISION OUTCOMES

3

ALLOW • REVIEW • BLOCK

ENVIRONMENTS

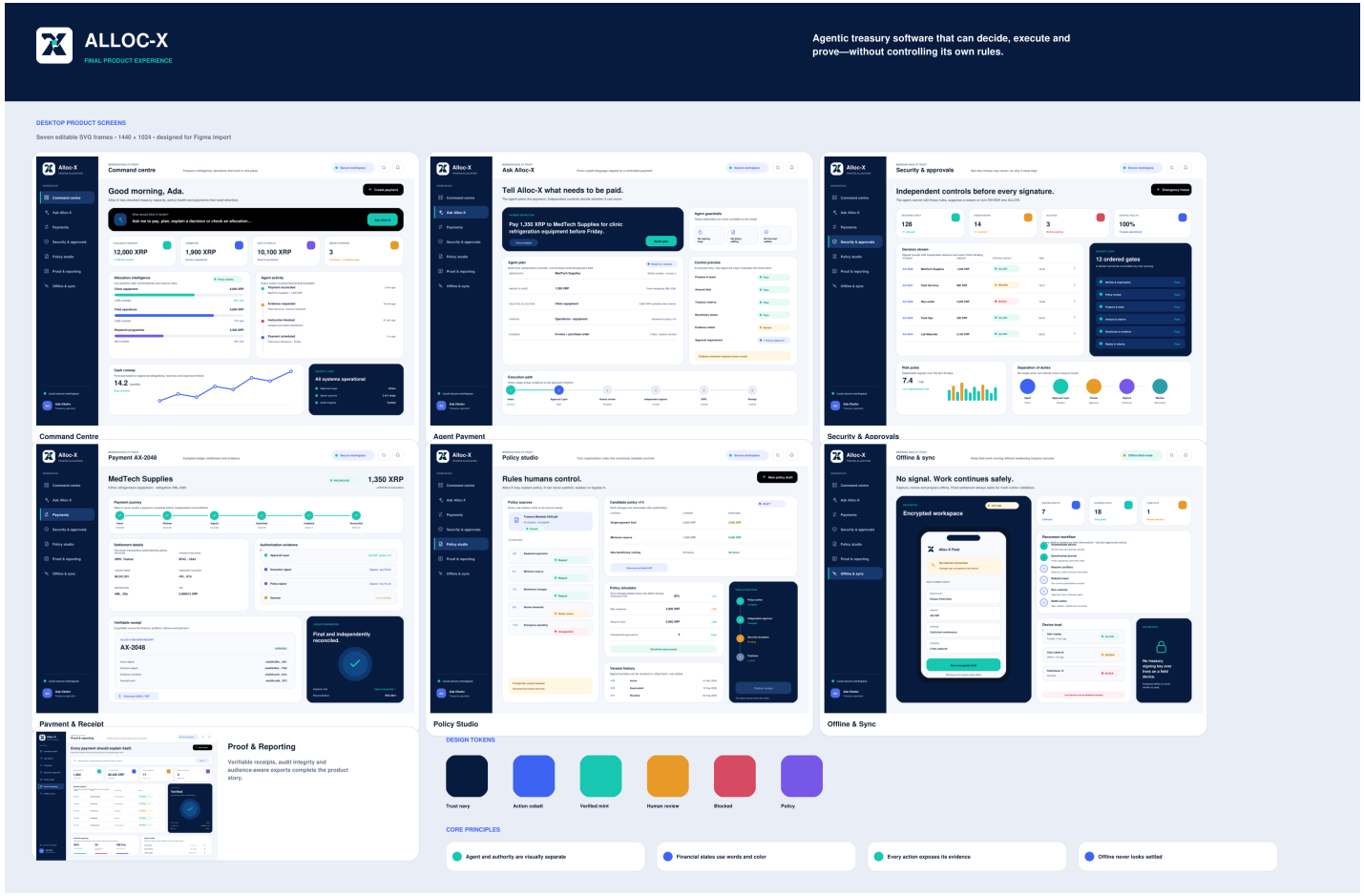
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Isolated release stages

Experience design • System architecture • Technology stack • Infrastructure • Environment strategy

Figma product experience

Seven connected product surfaces define how users command, control and verify the final Alloc-X experience.



FINAL PRODUCT SURFACES

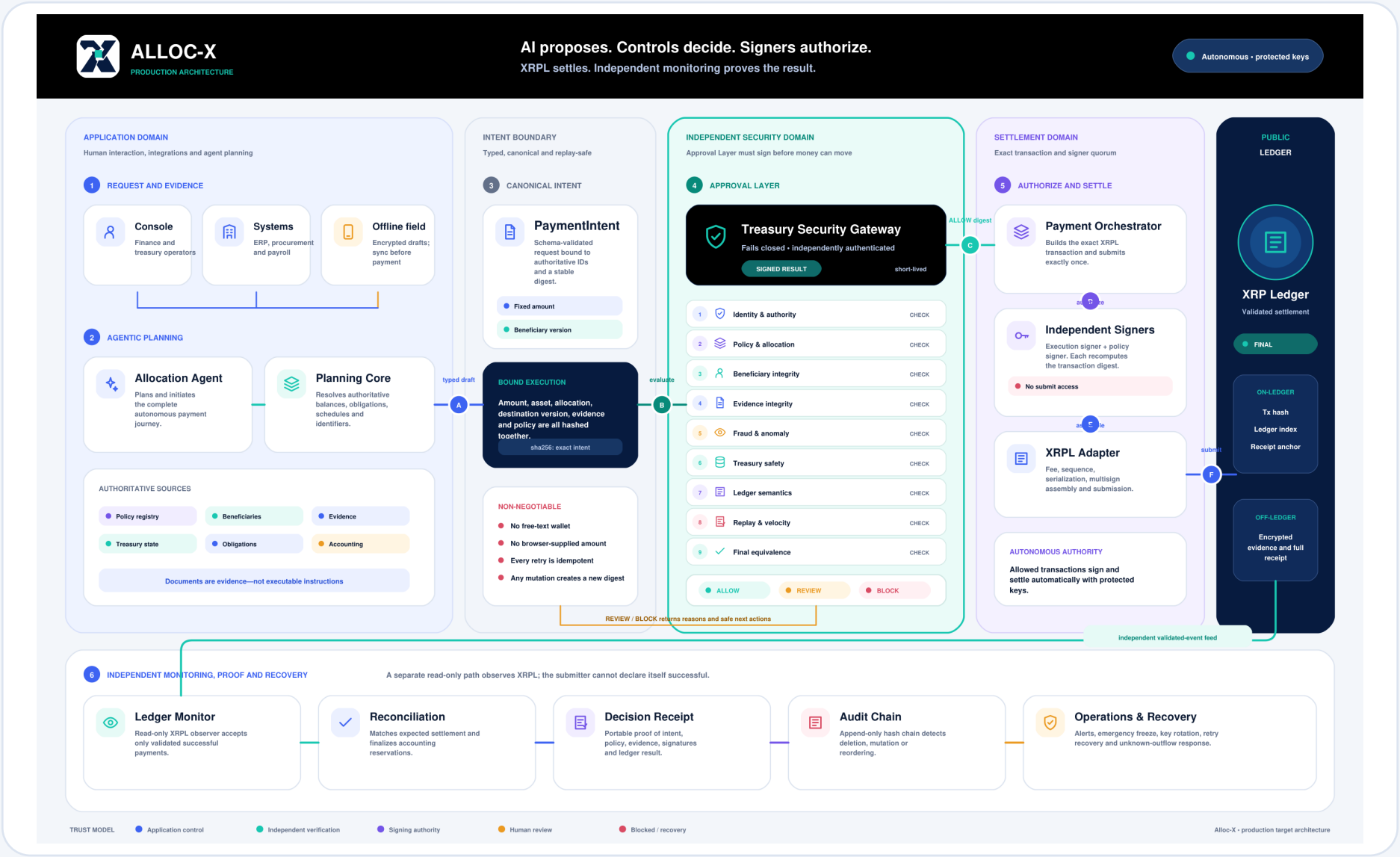
1. Command Centre
2. Ask Alloc-X
3. Security & Approvals
4. Payment & Receipt
5. Policy Studio
6. Offline & Sync
7. Proof & Reporting

DESIGN PRINCIPLES

- The agent and payment authority remain visually separate.
- Every payment state exposes its reason and evidence.
- Financial outcomes use unmistakable words and colors.
- Offline preparation is never displayed as settled.

Architecture: autonomy with protected authority

The agent initiates the complete journey; independent controls and protected signers make autonomous settlement safe.



What is to be implemented

The delivery plan is organized by authority boundary so each capability can be verified independently.

1 Agent command and planning

PLANNED

- Plain-language and voice payment instructions.
- Allocation selection, cash planning and evidence requests.
- Server-resolved payment intents with no browser-supplied economic fields.

2 Approval Layer and policy studio

PLANNED

- Ordered security gates with ALLOW, REVIEW and BLOCK outcomes.
- Versioned policies, simulation, publication and rollback.
- Beneficiary, invoice, reserve, velocity and anomaly controls.

3 Autonomous XRPL settlement

PLANNED

- Canonical transaction construction and exact-digest signing.
- Independent signer quorum with protected key custody.
- Submission, validation tracking and retry-safe orchestration.

4 Proof, offline operation and expansion

PLANNED

- Independent reconciliation, receipts and tamper-evident audit history.
- Encrypted offline drafts with conflict-aware synchronization.
- Configurable policy packs for nonprofit, commercial and public-sector users.

TARGET OUTCOME One user instruction becomes a controlled, signed, validated and independently reconciled payment.

Technology stack

The stack prioritizes typed contracts, deterministic controls, durable workflows and replaceable infrastructure adapters.

Product experience

Figma • React • Next.js • TypeScript • PWA

Responsive treasury console, accessible approval workflows, offline field capture and installable mobile experience.

Agent and application services

Node.js • TypeScript • JSON Schema • Tool contracts

Model-agnostic reasoning with bounded tools; authoritative identifiers and financial values resolved by server services.

Policy, data and workflows

PostgreSQL • Redis/Queue • Object storage • Workflow engine

Versioned policies, payment state machines, encrypted evidence, idempotency, leases, retries and reconciliation.

XRPL and cryptography

XRPL.js • Multi-signing • Ed25519 • SHA-256

Native-XRP settlement first, exact transaction binding, signer quorum, receipt anchoring and validated-ledger observation.

Identity and security

OIDC • MFA • RBAC/ABAC • KMS/HSM • WAF

Tenant isolation, step-up authentication, service identities, protected key custody, rate limits and emergency freeze.

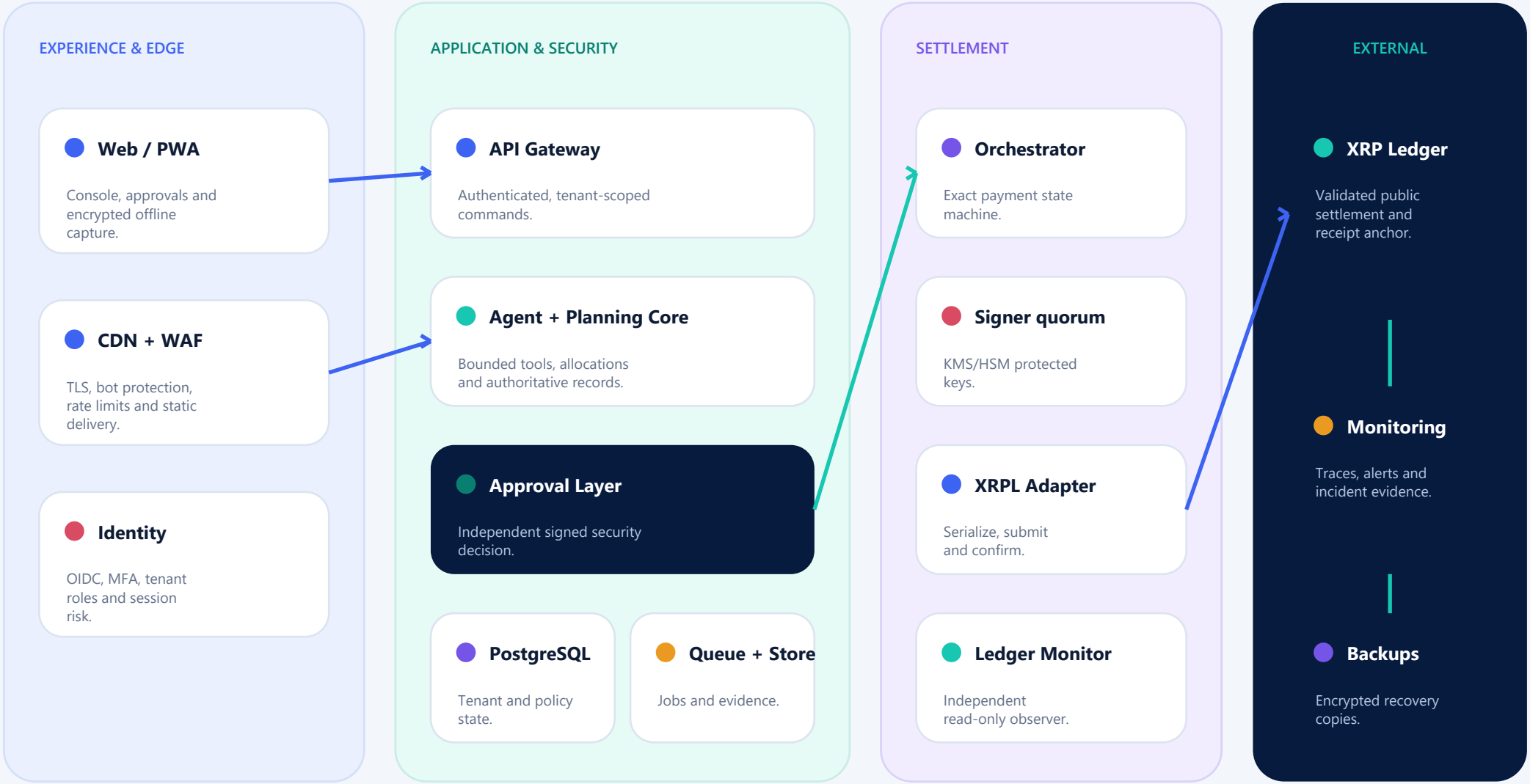
Delivery and observability

Docker • Terraform • CI/CD • OpenTelemetry • Sentry/Grafana

Repeatable environments, policy-as-code, release gates, distributed tracing, alerts and incident evidence.

Infrastructure and deployment topology

Security-critical services run with separate identities, credentials and network policies; no single runtime can spend alone.



Every environment receives separate databases, service identities, queues, buckets and signing keys.

Repository and environment configuration

A monorepo keeps shared contracts consistent while deployment boundaries preserve independent authority.

TARGET REPOSITORY STRUCTURE

```
apps/
  console/
  api/
  allocation-agent/
services/
  approval-layer/
  payment-orchestrator/
  signers/
  xrpl-adapter/
  ledger-monitor/
  receipt-builder/
  policy-registry/
packages/
  contracts/
  authorization/
  policy-engine/
  accounting/
  audit-chain/
  beneficiaries/
infra/
  environments/
  terraform/
  policies/
verification/
  integration/  security/  ledger/
```

ENVIRONMENT ISOLATION

DEVELOPMENT	Sandbox identities • settlement disabled by default
XRPL TESTNET	Faucet accounts • isolated keys • explicit release flag
STAGING	Production-like observability • no main-network keys
PRODUCTION	XRPL Mainnet • KMS/HSM • multi-person activation

CONFIGURATION CONTRACT

Runtime-injected secrets • never stored in source control

`APP_ENV`
deployment environment

`XRPL_NETWORK`
testnet | devnet | mainnet

`XRPL_ENDPOINT`
environment-scoped WebSocket

`DATABASE_URL`
secret-injected relational store

`QUEUE_URL`
durable workflow transport

`OBJECT_STORE_BUCKET`
encrypted evidence and receipts

`OIDC_ISSUER`
trusted identity provider

`KMS_KEY_ID`
protected signer key reference

`APPROVAL_PUBLIC_KEY`
decision verification key

`SETTLEMENT_ENABLED`
fail-closed activation flag